

Battery Fast-Charging Protocol Design via Large Language Model-Guided Multiobjective Bayesian Optimization

Anonymous Authors

Abstract—Fast, safe, and durable charging of lithium-ion batteries requires balancing charging speed, thermal stress, and degradation, which makes charging-protocol design a constrained multiobjective optimization problem. Bayesian optimization (BO) is attractive in this setting because it can approximate a Pareto set with a limited number of expensive simulator evaluations, but its early search is highly sensitive to the small initial design and to the absence of task-specific structure. This paper proposes LLMBO-MO, a large language model (LLM)-guided extension of a ParEGO-style decomposition-based multiobjective BO backbone that addresses this early-search gap. The proposed method introduces two LLM touchpoints that are kept strictly separable from the numerical surrogate: (i) an LLM-informed initialization that converts qualitative battery knowledge into a protocol-domain-valid and decision-space-diverse initial portfolio, and (ii) a weight-conditioned regional guidance mechanism that uses the active decomposition direction and the current optimization state to add region-informed restarts to early expected-improvement optimization, while the original numerical search is retained as an automatic fallback. All selected protocols are evaluated by the battery simulator before they enter the surrogate training set or the Pareto archive, and path-dependent voltage and temperature feasibility is determined by the simulator rather than by the LLM. Under 56 simulator evaluations and five seeds, LLMBO-MO attains a mean final hypervolume of 0.3848 ± 0.0073 on Chen2020, compared with 0.3520 ± 0.0169 for ParEGO, a 9.3% relative improvement; a similar advantage is observed on Ecker2015. Mechanism studies show that the gain is concentrated in the LLM-informed initialization, whereas regional guidance acts as a secondary early-search refinement. The conclusions are limited to simulation-based evaluation, a control-oriented degradation proxy, a single LLM backend, and matched-seed fixed-budget comparisons against a backbone-matched comparator set.

Index Terms—Battery charging, Bayesian optimization, large language models, lithium-ion batteries, multiobjective optimization.

I. INTRODUCTION

Fast and reliable charging is a central requirement for lithium-ion batteries used in electric vehicles and energy-storage systems. Increasing the charging current can shorten charging time, but it also intensifies heat generation and degradation, creating inherent trade-offs among charging speed, thermal safety, and battery durability [1]–[3]. Because the preferred compromise varies across operating scenarios, charging-protocol design is naturally a constrained multiobjective optimization problem whose desired output is a set of Pareto-nondominated protocols rather than a single schedule selected by fixed preferences.

Accurate evaluation of a charging protocol is costly. Physical cycling requires substantial time and laboratory resources, particularly when degradation is included. Physics-based

electrochemical–thermal simulation reduces reliance on repeated experiments [29], [30], but each candidate still requires the numerical solution of coupled battery dynamics. Consequently, the number of simulator evaluations becomes a major computational bottleneck. Bayesian optimization (BO) is well suited to this setting because a probabilistic surrogate and an acquisition function can prioritize informative evaluations under a limited budget [4]–[7]. For multiple objectives, decomposition-based multiobjective BO (MOBO) can sequentially explore different trade-off directions and approximate a Pareto set with relatively few expensive evaluations [8], [16], [20].

Existing studies improve BO-based charging design mainly by refining the search strategy or by incorporating additional task knowledge. Mesh-adaptive and parallel acquisition strategies improve the allocation of new evaluations [13], [15], whereas digital twins, aging models, electrochemical constraints, and related physics-based information provide task-specific numerical guidance [12], [14], [17]. These approaches are effective when suitable models or historical numerical information are available. They are less direct, however, for incorporating qualitative charging knowledge such as plausible multistage current patterns, SOC allocation heuristics, and thermal–degradation trade-offs, which is exactly the information that practitioners carry when they design a new charging protocol. This limitation matters most at the beginning of BO: with only a few evaluated protocols, the surrogate is highly sensitive to the initial design, and poor initial samples may consume a substantial fraction of a tight simulation budget before informative regions are reached.

Large language models (LLMs) provide a possible interface for translating such textual knowledge into optimization suggestions. LLAMBO introduced LLM-assisted warm starts, candidate generation, and surrogate-related reasoning for BO [23]. Subsequent studies have considered point or region preferences and selective use of inexpensive LLM predictions [24], [25]. LLM-assisted BO has also been applied to battery parameter identification [26], while broader studies show that the value of LLM guidance depends strongly on alignment with the target scientific domain [27]. For multiobjective fast-charging design, two aspects remain insufficiently explored. First, raw LLM suggestions must be converted into a protocol-domain-valid and diverse initial design before they can provide useful information to BO. Second, if LLM guidance is retained during sequential search, the recommendation should reflect the active multiobjective trade-off rather than remain independent of the current decomposition direction and optimization state. In

either case, LLM suggestions should guide *where* the optimizer searches, while simulator evaluations continue to determine objective values and path-dependent physical feasibility.

The closest prior work in fast-charging protocol design follows a related but distinct strategy. EIMO [10] transfers *optimization experience* from previously optimized state-of-health (SOH) levels to a new SOH level through a transfer acquisition function, and GPA-MOBO [9] accelerates a decomposition-based MOBO backbone using a Gaussian-process surrogate with multiple user-preference directions. Both target the same application and share the same decomposition-GP-EI numerical backbone as the present work. LLMBO-MO differs from them in the *source* of prior information: it queries an LLM, which requires only battery-domain text, rather than running EIMO at multiple SOH levels or training a problem-specific surrogate. The two families are therefore complementary—EIMO is preferable when historical optimization data are available at multiple SOH levels, whereas LLMBO-MO is preferable when only domain knowledge is available in textual form.

This paper proposes *LLMBO-MO*, which augments a ParEGO-style decomposition-based MOBO backbone with two LLM-guided search mechanisms. *The main contributions are organized around a single problem-solution chain*: each contribution fixes one specific gap in applying LLM guidance to early-stage multiobjective battery optimization. The first contribution is a *LLM-informed initialization* that converts battery-domain textual knowledge into a protocol-domain-valid and decision-space-diverse initial portfolio; this addresses the data-scarce stage before sequential BO begins. The second contribution is a *weight-conditioned regional guidance* mechanism that supplies additional restarts to early expected-improvement optimization and is positioned as a secondary early-search refinement, not as a peer to the initialization; this reflects the modest and inconclusive 3-seed ablation result reported in Section V. The third contribution is a fixed-budget numerical validation on two SPM_e parameterizations with matched seeds, which quantifies the combined mechanism and the relative contribution of each component.

The principal findings and insights are summarized as follows.

- A decomposition-based MOBO backbone, when seeded with a battery-domain-validated and decision-space-diverse LLM portfolio, produces a higher mean final hypervolume than the same backbone initialized with random or generic-prompt candidates under matched 26-evaluation seeds.
- The LLM-derived improvement is concentrated in the initialization step; the additional bounded regional guidance contributes a smaller and statistically inconclusive three-seed refinement.
- The advantage transfers to a second SPM_e parameterization but is interpreted as an additional battery-model check rather than as a formal cross-battery transfer study.
- The LLM is restricted to candidate locations; all objective values and Pareto-set entries are simulator-validated, which keeps the surrogate training set and the Pareto archive free of unevaluated LLM outputs.

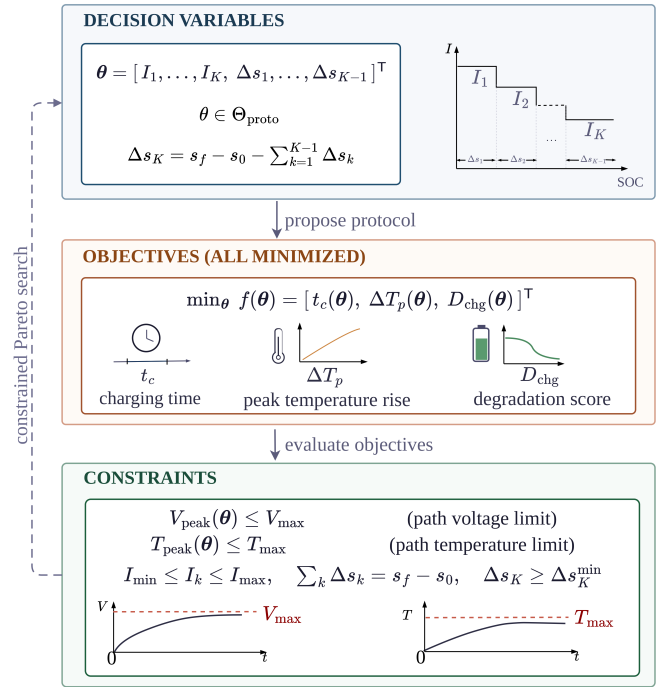


Fig. 1. Fast-charging multiobjective design problem.

Note to practitioners. LLMBO-MO is best viewed as an early-stage search aid when (i) the simulator is too expensive for a long BO budget, (ii) domain knowledge is available in textual form but not as a numerical prior, and (iii) the practitioner is willing to maintain a small LLM prompt and response log for reproducibility. The method adds at most $n_0 + T_L$ LLM calls to the optimization loop (six initialization queries plus a bounded early-window sequence in the present configuration), so its incremental API cost is modest compared with the simulator cost. If the LLM backend is unavailable, the LLM touchpoints degrade gracefully: the initialization falls back to a random protocol-domain-valid portfolio, and the regional-guidance branch is skipped, leaving the underlying ParEGO-style MOBO loop intact.

The rest of this paper is organized as follows. Section II formulates the multiobjective fast-charging problem and the MCC protocol geometry following Ref. [11]. Section III introduces the battery simulation model and the control-oriented degradation proxy. Section IV develops LLMBO-MO. Section V reports the numerical validation and discussion, and Section VI concludes the paper.

II. PROBLEM FORMULATION

The charging-design problem is formulated as the selection of a multistage constant-current protocol that balances charging speed, thermal safety, and degradation while satisfying operating constraints. Fig. 1 summarizes the protocol variables, optimization objectives, and feasibility conditions.

A. Multistage Constant-Current Protocol

A multistage constant-current (MCC) protocol divides charging into K stages. Stage k is characterized by a current

coordinate I_k and a delivered state-of-charge (SOC) span Δs_k . The battery is charged from an initial SOC s_0 to a target SOC s_f . The independent decision vector and the dependent terminal SOC span are

$$\begin{aligned}\theta &= [I_1, \dots, I_K, \Delta s_1, \dots, \Delta s_{K-1}]^\top, \\ \Delta s_K &= s_f - s_0 - \sum_{k=1}^{K-1} \Delta s_k.\end{aligned}\quad (1)$$

The admissible domain Θ_{proto} imposes box bounds on the currents and SOC spans and requires $\Delta s_K \geq \Delta s_K^{\min}$. This lower bound prevents a degenerate terminal stage and ensures that the protocol covers the prescribed SOC interval.

Following established BO-based charging formulations [7], [8], SOC spans rather than stage durations are optimized. If both current and duration were independent variables, the delivered charge would introduce a strongly coupled and potentially redundant representation. SOC-span coordinates instead place the switching boundaries directly in the SOC domain and simplify constraint handling. The experiments use $K = 3$, as in those studies.

Let $I_k^{\text{cell}} = \phi_{\text{cell}}(I_k)$ denote the physical cell current associated with the protocol coordinate I_k . The charging time is

$$t_c(\theta) = \kappa_t Q_{\text{eff}} \sum_{k=1}^K \frac{\Delta s_k}{I_k^{\text{cell}}}, \quad (2)$$

where Q_{eff} is the effective capacity and κ_t converts charge to time units.

B. Constrained Multiobjective Charging Design

The charging-design objectives quantify charging speed, thermal response, and degradation severity. For a protocol θ , the minimized objective vector contains charging time $t_c(\theta)$, peak temperature rise $\Delta T_p(\theta)$, and the protocol-level degradation proxy $D_{\text{chg}}(\theta)$. The latter two objectives are defined by the battery model in Section III. The constrained multiobjective problem is

$$\begin{aligned}\min_{\theta \in \Theta_{\text{proto}}} \quad & \mathbf{f}(\theta) = \begin{bmatrix} t_c(\theta) \\ \Delta T_p(\theta) \\ D_{\text{chg}}(\theta) \end{bmatrix}, \\ \text{s.t.} \quad & I_{\min} \leq I(t; \theta) \leq I_{\max}, \\ & s_0 \leq \text{SOC}(t; \theta) \leq s_f, \\ & V(t; \theta) \leq V_{\max}, \\ & T(t; \theta) \leq T_{\max}, \quad 0 \leq t \leq t_c(\theta).\end{aligned}\quad (3)$$

The protocol domain enforces the explicit current and SOC-coordinate bounds. The battery simulator checks the path-dependent voltage and temperature limits. A protocol that violates either limit or fails to reach s_f is infeasible and is excluded from the Pareto set.

The objectives are inherently conflicting. Higher currents can shorten charging time but generally increase heat generation and degradation severity. The preferred compromise varies across operating scenarios. The optimization target is therefore a feasible Pareto set that preserves these trade-offs rather than a single protocol selected using a fixed preference vector.

C. Expensive Black-Box Evaluation

For each candidate protocol, the battery model returns the objective vector and a feasibility indicator,

$$\mathcal{S}(\theta) \rightarrow (\mathbf{f}(\theta), \mathbb{I}_{\text{feas}}(\theta)). \quad (4)$$

Each call to $\mathcal{S}$ requires a coupled electrochemical–thermal simulation and evaluation of the degradation proxy described in Section III. The resulting optimization challenge is to approximate the feasible Pareto set with as few simulator calls as possible.

III. BATTERY SIMULATION MODEL

Each candidate protocol is evaluated by a common battery-model pipeline consisting of an electrochemical–thermal simulator and a protocol-level degradation proxy. Using the same evaluator for all optimization methods isolates differences caused by the search algorithms rather than by the objective model.

A. Electrochemical–Thermal Model

The electrochemical dynamics are represented by a Single Particle Model with electrolyte dynamics (SPMe). The SPMe retains solid-phase diffusion, electrolyte transport, and interfacial reaction kinetics at substantially lower computational cost than a full porous-electrode model [29]. The implementation uses PyBaMM [30]. A spatially lumped thermal model is coupled to the electrochemical simulation. The thermal balance and the reported temperature-rise objective are

$$\begin{aligned}C_{\text{th}} \frac{dT}{dt} &= \dot{Q}_{\text{gen}} - hA(T - T_{\text{amb}}), \\ \Delta T_p(\theta) &= \max_t T(t) - T_{\text{init}},\end{aligned}\quad (5)$$

where C_{th} is the effective thermal capacitance, $\dot{Q}_{\text{gen}}$ collects electrochemical, ohmic, and reversible heat, and hA is the lumped heat-transfer coefficient. Because the thermal model is spatially lumped, ΔT_p represents the peak cell-level temperature rise and does not resolve local temperature gradients or thermomechanical loading.

The Chen2020 parameter set is used for the LG INR21700-M50 case [31], while Ecker2015 provides a second battery parameterization [32]. Cell-specific capacity, voltage, and temperature limits are applied, while the protocol representation and optimization procedure remain unchanged. Successive charging stages are simulated sequentially so that the terminal electrochemical and thermal state of one stage initializes the next.

B. Protocol-Level Degradation Model

A single protocol simulation does not resolve long-horizon capacity fade. Accordingly, the third objective is a control-oriented degradation proxy rather than a direct prediction of

capacity lost during one charge. Let $\bar{s}$, $\bar{T}$, and $\bar{I}$ denote the time-averaged SOC, temperature, and current over the completed charging trajectory. The implementation evaluates

$$\chi(\bar{s}, \bar{T}, \bar{I}) = (a_s \bar{s} + b_s) \exp\left(\frac{\beta_I \bar{I} - E_a}{R \bar{T}}\right), \quad (6)$$

$$D_{\text{chg}}(\theta) = \kappa_D Q_{\text{eff}} \left(\frac{\chi(\bar{s}, \bar{T}, \bar{I})}{q_a} \right)^{1/\gamma}.$$

Here a_s , b_s , β_I , E_a , R , q_a , γ , and κ_D are fixed proxy coefficients shared by all optimization methods. The construction is related to the severity-factor family of control-oriented semi-empirical aging models [28].

The scale factor κ_D preserves the implemented objective scale; therefore D_{chg} is reported in arbitrary proxy units rather than as percentage capacity loss. Because the proxy uses trajectory averages, it does not separately resolve short high-SOC, high-current, or high-temperature excursions. This limitation is retained when interpreting the Pareto protocols.

In summary, the battery model maps each protocol θ to $\mathbf{f}(\theta)$ together with a feasibility flag. For BO, this mapping is treated as an expensive black box. Only simulator-evaluated objective values are used for surrogate fitting and Pareto-set construction.

IV. PROPOSED LLMBO-MO

The battery model in Section III maps each candidate protocol to an objective vector together with a feasibility flag, so the design problem is an expensive black-box evaluation for BO. LLMBO-MO interfaces with this black box at a single point: the candidate-protocol decision vector θ . The LLM is allowed to read the battery-domain text, the decision-variable bounds, and the current BO state (active decomposition weight, observed objective ranges, selected evaluated protocols), but it does not have access to the simulator, the GP surrogate, the acquisition surface, or the feasible Pareto archive. All objective values entering the surrogate training set and the Pareto archive are produced by the simulator, not by the LLM.

The following subsection motivates the design of LLMBO-MO from the limitations of the closest prior work; the numerical backbone and the two LLM mechanisms are then described in detail.

A. Overview

LLMBO-MO augments a ParEGO-style decomposition-based multiobjective Bayesian optimization backbone with two LLM-guided search mechanisms. The numerical MOBO loop remains responsible for trade-off decomposition, surrogate fitting, acquisition optimization, simulator evaluation, and Pareto updating. The LLM interacts with this loop only through candidate locations. Before sequential BO, it contributes to a domain-informed initial design. During a bounded early-search window, it may additionally provide a weight-conditioned region preference that supplies extra restarts to the numerical acquisition optimizer. The original numerical search remains available in both stages, and no unevaluated LLM proposal is used as a surrogate target or Pareto value. The complete workflow is summarized in Fig. 2.

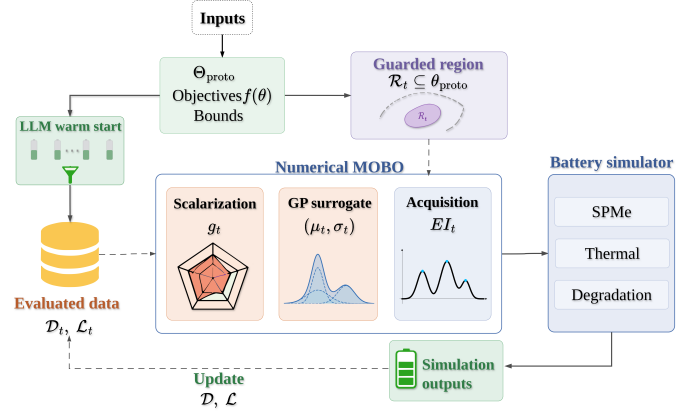


Fig. 2. Structure of LLMBO-MO

B. Design Rationale and Motivation

The design of LLMBO-MO is guided by four explicit considerations that together distinguish the proposed method from the closest prior work.

Why an LLM rather than a rule-based template, a fine-tuned prior model, or a full historical-archive transfer. Ref. [10] transfers *optimization experience* across state-of-health levels and therefore requires running the algorithm at multiple SOH levels before the target SOH is reached. Ref. [9] trains a Gaussian-process surrogate from past runs. Both approaches are effective when the necessary historical optimization data are available, but they require data that are not always present in a new battery project. LLMBO-MO instead queries an LLM, which needs only battery-domain text, so the same prompt template is portable across cell chemistries, cell formats, and operating windows with only the context block being replaced. A hand-coded multi-stage template generator (a *rules* baseline) is also a possible source of qualitative information, and the present manuscript explicitly includes such a comparator in the initialization study to disentangle prompt quality from domain knowledge.

Why exactly two LLM touchpoints rather than a single end-to-end interface. A single LLM call embedded inside the BO loop would conflate *where to look* with *what the objective is* and would require trusting the LLM with surrogate updates or Pareto assignments, which violates the LLM-to-surrogate boundary that the present work enforces. Two narrowly scoped touchpoints keep the LLM strictly on the candidate side of the boundary: one touchpoint at initialization (data-scarce stage) and one touchpoint at early acquisition (where the surrogate is still uncertain).

Why initialization plus a bounded early window rather than full-horizon guidance. The prompt study in Section V shows that the LLM contribution is largest at initialization and decreases as the surrogate accumulates task-specific data. Restricting the regional guidance to an early window $t \leq T_L$ therefore concentrates the LLM call budget where it is most informative and avoids the LLM and the GP surrogate competing for the same decision during the data-rich stage. The early-window restriction is also what makes the API cost predictable and what allows the LLM to be replaced by a

random-region baseline for robustness checks.

Why weight-conditioning rather than objective-conditioning or hypervolume-conditioning. In a decomposition-based MOBO, the active trade-off weight $w^{(t)}$ fully specifies the current scalarization of the problem, and conditioning the LLM on $w^{(t)}$ directly aligns the regional preference with the scalarized target. Objective-conditioning would have to aggregate three competing objectives into a single LLM input, and hypervolume-conditioning would have to communicate a set-valued signal that is harder for an LLM to reason about. The chosen design therefore uses the same Das–Dennis-simplex weight schedule that the underlying ParEGO-style backbone uses, which keeps the LLM interface and the numerical interface synchronized at every iteration.

Taken together, these four considerations motivate a design that (a) treats the LLM as a portable domain-knowledge interface, (b) keeps the LLM strictly on the candidate side of the LLM-to-surrogate boundary, (c) restricts the LLM call budget to the data-scarce and early-uncertainty stages, and (d) synchronizes the LLM interface with the decomposition weight used by the numerical backbone.

C. Decomposition-Based Multiobjective BO Backbone

At iteration t , LLMBO-MO selects a trade-off weight $w^{(t)}$ from a set $\mathcal{W}$ initialized on a Das–Dennis simplex lattice and dispersed by projected Riesz-energy relaxation [33]. Because the three objectives have different numerical ranges, charging time and the degradation proxy are first logarithmically transformed,

$$\begin{aligned}\tilde{\mathbf{f}}(\theta) &= [\ell_t(\theta), \Delta T_p(\theta), \ell_D(\theta)]^\top, \\ \ell_t(\theta) &= \log_{10} \max(t_c/t_{\text{ref}}, \epsilon_t), \\ \ell_D(\theta) &= \log_{10} \max(D_{\text{chg}}/D_{\text{ref}}, \epsilon_D).\end{aligned}\quad (7)$$

For objective i , let $R_{i,t}^{\text{hist}}$ denote its observed range and $R_i^{\text{glob}} = \bar{y}_i^{\text{ref}} - \bar{y}_i^{\text{ideal}}$. The implementation uses

$$s_{i,t} = \begin{cases} R_i^{\text{glob}}, & R_{i,t}^{\text{hist}} < \rho_R R_i^{\text{glob}}, \\ R_{i,t}^{\text{hist}}, & \text{otherwise,} \end{cases}\quad (8)$$

which prevents a narrow early observed range from dominating the scalarization.

The transformed objectives are decomposed through an augmented Tchebycheff function,

$$\begin{aligned}\tilde{f}_{i,t}(\theta) &= \frac{|\tilde{f}_i(\theta) - \tilde{z}_{i,t}^*|}{s_{i,t}}, \\ g_t(\theta | w^{(t)}) &= \max_i w_i^{(t)} \tilde{f}_{i,t}(\theta) + \rho_{\text{aug}} \sum_{i=1}^m w_i^{(t)} \tilde{f}_{i,t}(\theta),\end{aligned}\quad (9)$$

where $w^{(t)} \geq \mathbf{0}$, $\mathbf{1}^\top w^{(t)} = 1$, and $m = 3$. All feasible observations are re-scalarized for the current weight at every iteration, so one numerical BO loop can sequentially explore different parts of the Pareto front.

An ARD Matérn-5/2 GP with a white-noise term is fitted to standardized scalar targets on normalized decision variables.

Let $\mu_t(\theta)$ and $\sigma_t(\theta)$ denote the GP posterior mean and standard deviation. For minimization, EI is

$$\begin{aligned}z_t(\theta) &= \frac{g_t^{\min} - \mu_t(\theta)}{\sigma_t(\theta)}, \\ \text{EI}_t(\theta) &= (g_t^{\min} - \mu_t(\theta))\Phi(z_t) + \sigma_t(\theta)\phi(z_t),\end{aligned}\quad (10)$$

where $g_t^{\min}$ is the best observed scalar value and $\Phi(\cdot)$ and $\phi(\cdot)$ are the standard normal cumulative distribution and density functions. The acquisition optimizer uses the incumbent, Gaussian and uniform candidates, and multi-start L-BFGS-B solutions. The selected protocol is repaired to Θ_{proto} , evaluated by $\mathcal{S}$, appended to $\mathcal{D}$, and used to update the feasible Pareto archive $\mathcal{P}$. This ParEGO-style scalarization–GP–EI loop [20] is the numerical backbone shared with the primary BO comparator.

D. LLM-Informed Initialization

The first LLM mechanism targets the data-scarce stage before sequential BO begins. Its prompt specifies the battery context, decision-variable bounds, three-stage MCC geometry, objective definitions, and qualitative charging knowledge such as reasonable current ordering and SOC allocation. The LLM returns a pool of numerical protocol candidates together with a heuristic ranking score that is used only during candidate selection. Non-numerical or non-finite entries and duplicates are removed, box violations are clipped, and candidates that cannot satisfy $\Delta s_K \geq \Delta s_K^{\min}$ are rejected. These operations enforce protocol-domain admissibility only; voltage and temperature feasibility are still determined by the simulator.

Let $h_{\text{ws}}(\theta) \in [0, 1]$ denote the LLM-provided heuristic score. A soft penalty discourages a terminal SOC span below a preferred boundary,

$$p_{\text{soft}}(\theta) = \text{clip}\left(\frac{\Delta s_K^{\text{soft}} - \Delta s_K}{w_{\text{soft}}}, 0, 1\right).\quad (11)$$

Let $\mathbb{I}_{\text{mono}}(\theta)$ indicate a nonincreasing current sequence. Candidate selection combines the LLM score, the soft protocol heuristic, and decision-space diversity,

$$\begin{aligned}J_{\text{ws}}(\theta) &= h_{\text{ws}}(\theta) - \omega_p p_{\text{soft}}(\theta) + \omega_m \mathbb{I}_{\text{mono}}(\theta) \\ &\quad + \omega_d d_{\min}(\theta, \mathcal{C}_{\text{sel}}),\end{aligned}\quad (12)$$

where

$$d_{\min}(\theta, \mathcal{C}_{\text{sel}}) = \begin{cases} 0, & \mathcal{C}_{\text{sel}} = \emptyset, \\ \min_{\mathbf{c} \in \mathcal{C}_{\text{sel}}} \|\hat{\theta} - \hat{\mathbf{c}}\|_2, & \text{otherwise.} \end{cases}\quad (13)$$

The hat denotes normalized decision coordinates. Random protocol-domain-valid points complete the initial design so that the search is not restricted to LLM proposals. Every selected initial protocol is evaluated by the simulator before entering $\mathcal{D}$. Thus, the LLM changes the distribution of initial evaluation locations without introducing unevaluated objective values into the BO database.

Algorithm 1 LLMBO-MO

Require: Simulator $\mathcal{S}$, LLM $\mathcal{L}$, total budget N , initial budget n_0 , weight set $\mathcal{W}$, guidance horizon T_L
Ensure: Evaluated feasible Pareto set $\mathcal{P}$

- 1: Construct LLM-informed candidate pool; repair and rank using (12)
- 2: Combine selected LLM candidates with random protocol-domain-valid points
- 3: Evaluate n_0 initial protocols using $\mathcal{S}$ and initialize $\mathcal{D}, \mathcal{P}$
- 4: **for** $t = 1$ to $N - n_0$ **do**
- 5: Select trade-off weight $w^{(t)} \in \mathcal{W}$
- 6: Transform and scalarize feasible observations by (7)–(9)
- 7: Fit the GP and construct EI by (10)
- 8: **if** $t \leq T_L$ and regional guidance is enabled **then**
- 9: Query $\mathcal{L}$ using $w^{(t)}$ and the current BO state
- 10: Add anchors from an admissible region as EI-search restarts
- 11: **end if**
- 12: Maximize EI and select θ_{t+1}
- 13: Evaluate $\mathcal{S}(\theta_{t+1})$ and append the result to $\mathcal{D}$
- 14: Update the feasible Pareto archive $\mathcal{P}$
- 15: **end for**
- 16: **return** $\mathcal{P}$

E. Bounded Weight-Conditioned Regional Guidance

The second LLM mechanism operates only during an early window $t \leq T_L$ and refines acquisition optimization after an initial simulator-validated database has been formed. At iteration t , the prompt includes the current decomposition weight $w^{(t)}$, observed objective ranges, current best values, selected evaluated protocols, and a compact summary of the optimization state. The LLM can return a point, a bounded hyperrectangular region, or an explicit no-preference response. A point is converted to a local region with fixed-width margins; returned regions are parsed, clipped, repaired to Θ_{proto} , and rejected when they are too narrow, too broad, or otherwise inadmissible.

Anchors sampled from an accepted region are used only as additional restarts for the numerical EI maximization in (10); the original acquisition restarts and the full protocol domain are retained. Regional guidance therefore biases the numerical search without restricting the feasible search space or modifying objective values, GP training targets, the posterior model, or Pareto relations. Invalid or absent guidance simply leaves the original acquisition search unchanged, and all LLM regional interaction stops after T_L . The two mechanisms consequently play different roles: LLM-informed initialization provides a diversified global starting design, whereas regional guidance supplies a local, weight-conditioned search bias during the early sequential stage. The optional posterior-mean coupling studied during method development is not part of the principal algorithm and is retained only in Appendix A.

F. Algorithm Summary

Algorithm 1 summarizes the complete procedure. The numerical MOBO backbone remains active throughout the full budget, while LLM assistance is concentrated in the data-scarce initialization and early acquisition stages.

V. NUMERICAL VALIDATION AND DISCUSSION

A. Experimental Setup

The numerical study examines two questions: whether LLMBO-MO improves fixed-budget multiobjective charging

TABLE I
BATTERY AND SIMULATOR SETTINGS

Parameter	Value
Primary parameterization	Chen2020 (LG INR21700-M50)
Additional parameterization	Ecker2015
Nominal capacity (Chen2020)	5.0 Ah
Cell voltage range (Chen2020)	2.50–4.20 V
Cell format (Chen2020)	21700 cylindrical (70 × 21 mm)
Initial/ambient temperature	298.15 K
Maximum cell temperature	328.15 K

TABLE II
LLMBO-MO PARAMETER SETTINGS

Parameter	Value
Number of candidate solutions	15
Number of templates for generating solutions	2
Number of iterations for optimization	50
Number of initial samples	6
Number of trials	5
Engine used for chat-based tasks	V4-Flash
Percentage of top candidates to select	0.2
Soft-penalty weight ω_p	0.20
Monotonicity weight ω_m	0.10
Diversity weight ω_d	0.30
Preferred terminal SOC span Δs_K^{soft}	0.10
Soft-penalty width w_{soft}	0.10
Augmented Tchebycheff coefficient ρ_{aug}	0.05
Objective-scale ratio ρ_R	0.30
GP noise floor	1.0e-6

design, and which of the two LLM-guided search mechanisms accounts for the observed improvement. The SPMc-based simulator in Section III is evaluated with the Chen2020 and Ecker2015 parameterizations [31], [32]. Chen2020 is the primary case for algorithm comparison and mechanism studies; Ecker2015 provides an additional parameterization for checking whether the observed behavior is confined to one battery model. Each candidate protocol is evaluated in terms of charging time t_c , peak temperature rise ΔT_p , and degradation proxy D_{chg} .

The main comparisons use six initialization evaluations followed by 50 sequential BO evaluations, giving a total budget of 56 simulator calls. For each battery parameterization, the same five seeds (8409–8413) are used for every compared method. Hypervolume (HV) is the principal multiobjective performance metric because it jointly reflects convergence and coverage of the nondominated set; higher HV indicates a better solution set under a common reference convention. Within each parameterization, all methods use the same objective definitions, HV convention, and simulator budget. The battery and LLMBO-MO settings are summarized in Tables I and II, respectively. The listed cell specifications apply to the primary Chen2020 case; Ecker2015 uses its source parameterization. Reproducibility details are provided in Appendix C.

ParEGO is the primary BO comparator because it is a standard scalarization-based method for expensive multiobjective optimization and shares the decomposition–GP structure used by the numerical backbone of LLMBO-MO [20]. DISK and PIMD provide recent surrogate-assisted references for expen-

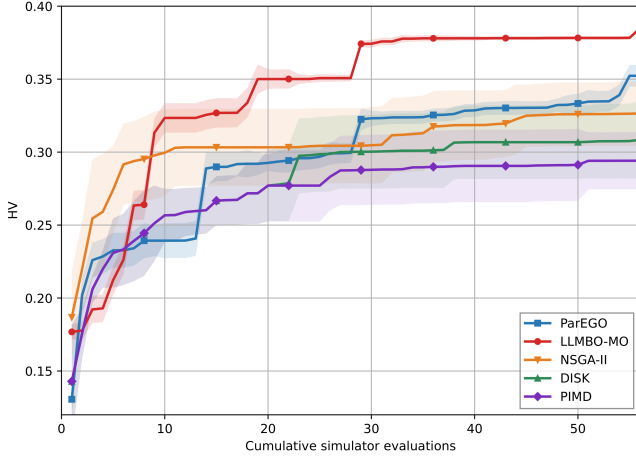


Fig. 3. Chen2020 hypervolume progress for LLMBO-MO and four multiobjective comparators under the 56-evaluation budget. Curves summarize the same five fixed seeds (8409–8413) for each method; shaded bands indicate one sample standard deviation.

TABLE III
CHEN2020 FINAL HYPERVOLUME OVER FIVE SEEDS

Method	Final HV (mean $\pm$ SD) $\uparrow$
LLMBO-MO	0.3848 ± 0.0073
ParEGO	0.3520 ± 0.0169
NSGA-II	0.3265 ± 0.0220
DISK	0.3085 ± 0.0267
PIMD	0.2941 ± 0.0195

sive multiobjective optimization [21], [22], while NSGA-II is included as a classical evolutionary multiobjective baseline [34]. Together, these methods provide both backbone-matched and complementary multiobjective references.

B. Overall Multiobjective Optimization Performance

The first study compares all methods under the common 56-call budget. Figure 3 presents the Chen2020 HV progress, and Table III reports the five-seed final statistics.

LLMBO-MO attains the highest mean final HV in the Chen2020 comparison. After 56 simulator evaluations, its five-seed mean is 0.3848 ± 0.0073 , compared with 0.3520 ± 0.0169 for ParEGO, corresponding to a 9.3% relative improvement in mean final HV. A paired Wilcoxon signed-rank test on the five matched seeds gives $p = 0.0625$ (two-sided), the 95% confidence interval for the mean difference obtained from the reported mean and SD with the $t(4)$ critical value is $[-0.0183, +0.0839]$, and Cohen's d_z lies between 1.78 (uncorrelated seeds) and 3.04 (correlated seeds). Because the 95% CI brackets zero under the conservative seed-correlation assumption, the Chen2020 result is reported as a *descriptive* improvement: the direction is consistent across seeds and across both parameterizations, but a five-seed test does not by itself establish statistical significance. The advantage emerges primarily during the early-to-middle portion of the budget and is retained as the numerical GP-EI loop proceeds. This behavior is consistent with the intended role of LLM guidance:

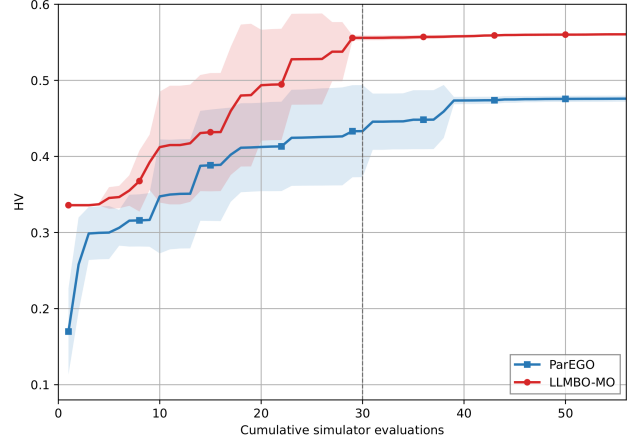


Fig. 4. Ecker2015 HV progress over the same five seeds (8409–8413). Lines show five-seed mean HV and bands show one sample standard deviation. The dashed line marks evaluation 30.

TABLE IV
ECKER2015 HV AT INTERMEDIATE AND FINAL BUDGETS

Method	Evaluation 30	Evaluation 56
ParEGO	0.4333	0.4760 ± 0.0039
LLMBO-MO	0.5559	0.5605 ± 0.0008

improving the allocation of scarce early simulator evaluations while leaving objective evaluation, surrogate fitting, and Pareto updating numerical. LLMBO-MO also achieves higher five-seed mean final HV than NSGA-II, DISK, and PIMD under the same evaluation budget.

The Ecker2015 experiment provides a second-parameterization check. The same 56-evaluation budget and the same five seeds are used. Figure 4 and Table IV show a similar performance separation.

At evaluation 30, the five-seed mean HV is 0.5559 for LLMBO-MO and 0.4333 for ParEGO. At evaluation 56, the corresponding means are 0.5605 ± 0.0008 and 0.4760 ± 0.0039 . A paired Wilcoxon signed-rank test on the five matched seeds gives $p = 0.0625$ (two-sided); the 95% confidence interval is $[-0.0017, +0.1707]$ under the $t(4)$ critical value, and Cohen's d_z lies between 5.81 and 21.6. Thus, the advantage observed on Chen2020 is also present under the Ecker2015 parameterization, although this experiment is interpreted as an additional battery-model check rather than a formal cross-battery transfer study, and the same descriptive (not inferential) caveat applies.

C. Effectiveness of the LLM-Guided Search Mechanisms

1) *LLM-Informed Initialization*: The first mechanism study isolates the effect of battery-specific textual knowledge during initialization. A separate Chen2020 experiment uses ten matched seeds and a shorter 26-evaluation budget. All variants share the same downstream numerical BO procedure and differ only in the source of the initial candidate portfolio.

Battery-specific LLM initialization improves mean final HV over random initialization by 0.05828. The detailed generic

TABLE V
CHEN2020 INITIALIZATION STUDY OVER TEN MATCHED SEEDS

Initialization source	Final HV $\uparrow$
Random	0.31062 $\pm$ 0.04303
Detailed generic prompt	0.31371 $\pm$ 0.04675
Battery-specific LLM prompt	0.36890 $\pm$ 0.00824

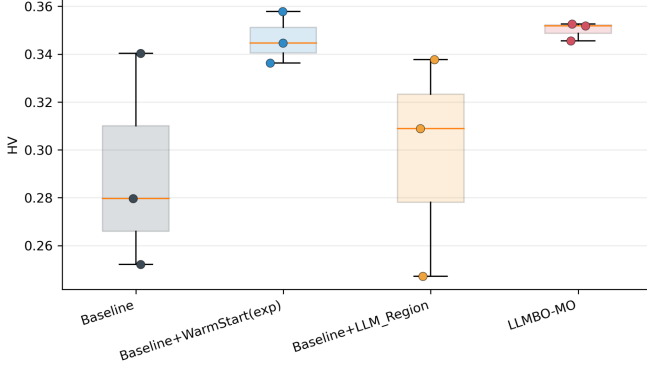


Fig. 5. Chen2020 short-budget component ablation over three matched seeds (8409–8411). Boxplots summarize final HV after 16 simulator evaluations.

prompt remains close to the random condition, whereas the battery-specific prompt yields a higher mean with substantially lower dispersion. The paired comparison remains supported after Holm multiplicity correction ($p = 0.005859$, $d_z = 1.309$). These results suggest that the observed gain is associated with battery-specific prompt content rather than with prompting alone, supporting the use of LLM-informed initialization in the data-scarce stage of MOBO.

2) *Bounded Regional Guidance*: The second mechanism study evaluates regional guidance both alone and in combination with LLM-informed initialization. The short-budget Chen2020 ablation uses three matched seeds (8409–8411), six initialization evaluations, and ten sequential BO evaluations, giving 16 simulator calls per seed. All four variants share the same numerical multiobjective BO backbone.

LLM-informed initialization produces the dominant short-budget improvement, raising mean HV from 0.2907 to 0.3462. Regional guidance alone changes the mean only modestly (0.2907 to 0.2980), and the three-seed experiment does not establish an independent systematic benefit for this component: a paired Wilcoxon signed-rank test on the three matched seeds gives $p = 0.875$ (two-sided) for the *baseline* vs. *region-only* comparison, $p = 0.125$ for the *baseline* vs. *LLM-informed initialization* comparison, and $p = 0.125$ for the *baseline* vs. *complete LLMBO-MO* comparison. Because the three-seed sample size does not give the Wilcoxon test adequate power at conventional α levels, these p values are reported for transparency rather than as evidence of significance; the descriptive mean differences should be read together with the dispersion columns of Table VI. When both mechanisms are enabled, however, the complete LLMBO-MO variant achieves the highest observed mean HV (0.3500) and the smallest observed dispersion (± 0.0039) among the four ablation variants. This result is consistent with a complementary role for regional

guidance: LLM-informed initialization supplies the main global improvement, while weight-conditioned region information can refine the subsequent early acquisition search.

D. Pareto Charging-Protocol Analysis

HV summarizes optimization quality, but the practical output of the optimizer is a set of charging protocols. Figure 6 and Table VII therefore examine representative simulator-evaluated nondominated solutions and their charging trajectories.

The representative protocols exhibit the expected physical trade-offs. More aggressive current schedules shorten the charging horizon but generally produce larger temperature excursions and higher degradation-proxy values. *Protocol A* (6.000/5.000/3.000 A, $\Delta s = 0.400/0.300/0.100$) sits at the speed-favouring corner of the Pareto front: the first two stages operate near the current upper bound, which drives $t_c = 2880$ s but also produces the largest peak temperature rise ($\Delta T_p = 7.567$ K) and the largest degradation-proxy value ($D_{\text{chg}} = 1.261$). The ΔT_p rise is dominated by the high current in the first two stages, whereas the D_{chg} rise is driven jointly by the high current and the high average SOC at which this high current is delivered. *Protocol E* (2.000/2.000/2.000 A, $\Delta s = 0.245/0.153/0.403$) sits at the durability-favouring corner: a constant current below the 3 A lower edge of the upper bracket, with the bulk of the SOC span allocated to the third stage, gives $t_c = 7200$ s but reduces ΔT_p to 1.528 K and D_{chg} to 0.640. *Protocols B–D* trace the intermediate trade-off, in which the current sequence is reshaped from the high-to-low staircase of A toward the more uniform schedule of E; the deformation of the current sequence is the dominant physical lever, whereas the SOC allocation shifts play a secondary role in this regime. Gentler schedules move toward the lower-temperature and lower-degradation regions of the Pareto front at the cost of longer charging time. The time-domain trajectories in Fig. 6 provide the corresponding voltage, temperature, current, and SOC evolution for these different operating compromises.

E. Discussion

The discussion interprets the numerical results in light of the four design considerations stated in Section IV-B.

1) *Summary of the main findings*: The main Chen2020 and Ecker2015 comparisons both show that LLMBO-MO achieves a higher five-seed mean final HV than the backbone-matched ParEGO under the same 56-evaluation budget, and the initialization study isolates the LLM contribution to the data-scarce stage. The advantage emerges in the early-to-middle portion of the budget and is retained as the numerical GP–EI loop proceeds, which is consistent with the intended role of LLM guidance: improving the allocation of scarce early simulator evaluations while leaving objective evaluation, surrogate fitting, and Pareto updating numerical. LLMBO-MO also achieves a higher five-seed mean final HV than NSGA-II, DISK, and PIMD under the same evaluation budget.

2) *Mechanism interpretation*: The 10-seed prompt study provides the strongest mechanism-level evidence: the battery-specific LLM prompt raises mean final HV by 0.05828 over random initialization and remains supported after Holm

TABLE VI
CHEN2020 SHORT-BUDGET COMPONENT ABLATION OVER THREE MATCHED SEEDS

Variant	Seed 8409	Seed 8410	Seed 8411	Mean $\pm$ SD $\uparrow$
Numerical MOBO baseline	0.2522	0.3404	0.2797	0.2907 ± 0.0451
+ LLM-informed initialization	0.3446	0.3363	0.3579	0.3462 ± 0.0109
+ Regional guidance only	0.2472	0.3377	0.3090	0.2980 ± 0.0462
Complete LLMO-MO	0.3456	0.3518	0.3527	0.3500 ± 0.0039

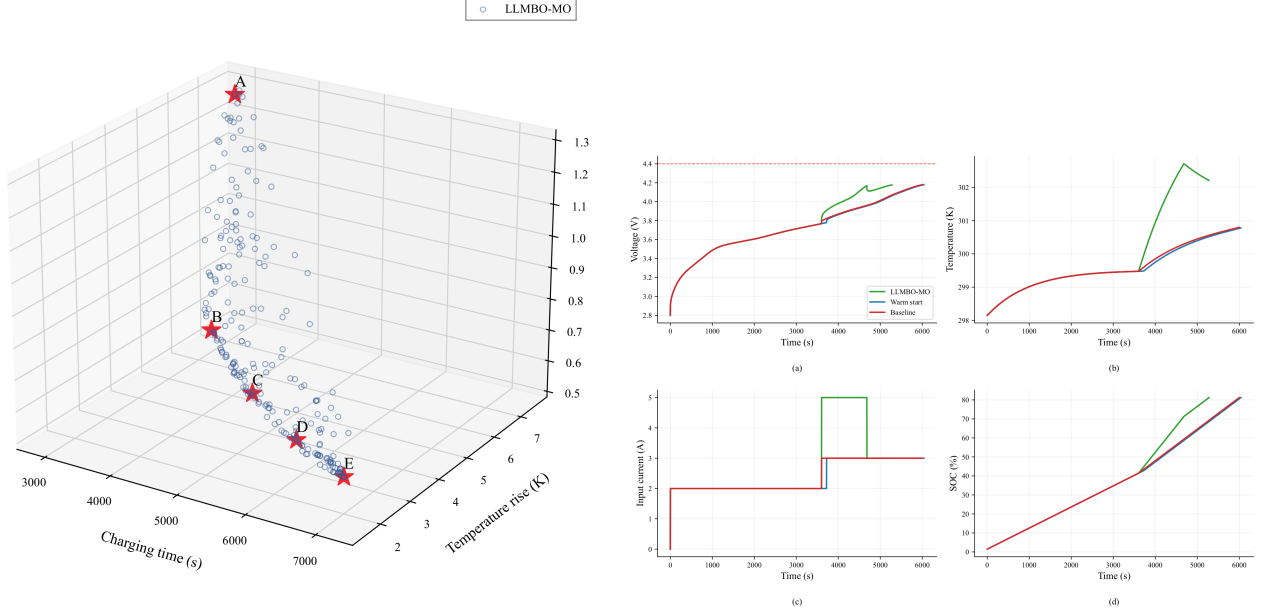


Fig. 6. Chen2020 Pareto-protocol interpretation. *Left*: feasible evaluated LLMO-MO protocols with representative nondominated solutions A–E. *Right*: simulated current, voltage, temperature, and SOC trajectories for selected LLMO-MO and comparison protocols.

TABLE VII
REPRESENTATIVE CHEN2020 LLMO-MO PARETO PROTOCOLS

Protocol	$I_1/I_2/I_3/\Delta s_1/\Delta s_2/\Delta s_3$	t_c (s)	ΔT_p (K)	D_{chg} (a.u.)
A	6.000/5.000/3.000/0.400/0.300/0.100	2880	7.567	1.261
B	3.475/3.633/3.000/0.100/0.219/0.481	4490	3.292	0.826
C	2.782/2.000/3.000/0.211/0.100/0.489	5199	3.069	0.680
D	2.000/3.942/3.000/0.400/0.100/0.300	5857	3.049	0.573
E	2.000/2.000/2.000/0.245/0.153/0.403	7200	1.528	0.640

The packed parameter order is three stage currents in amperes (A) followed by the three SOC spans. Each row is a simulator-evaluated feasible nondominated record, and the SOC spans sum to 0.8.

multiplicity correction ($p = 0.005859$, $d_z = 1.309$), whereas the detailed generic prompt remains close to the random condition. This pattern is consistent with the design rationale that the LLM contribution is largest at initialization and depends on the alignment between the prompt content and the battery domain. The 3-seed component ablation is statistically inconclusive for the regional-guidance branch alone, which is reported honestly in Section V. The complete LLMO-MO variant nevertheless achieves the highest observed mean HV and the smallest observed dispersion among the four short-budget variants, which is consistent with a complementary role for regional guidance: LLM-informed initialization supplies the main global improvement, while weight-conditioned region information can refine the subsequent early acquisition search.

3) *Physical trade-offs in the representative protocols*: The representative protocols in Table VII exhibit three physically distinct operating compromises. *Protocol A* is the speed-favouring corner (high current, high ΔT_p , high D_{chg}). *Protocol E* is the durability-favouring corner (uniform low current, low ΔT_p , low D_{chg}). The intermediate protocols (B–D) trace a continuous deformation in which the high-to-low staircase of A is progressively flattened toward the uniform schedule of E, and the SOC allocation shifts move the heat-generation load from the first stage to the third stage. The direction of this deformation is consistent with the LLM initialization portfolio: the early-search pool contains predominantly staircase-shaped protocols, and the GP–EI loop explores the neighbourhood of these staircase anchors before discovering the more uniform

schedules that ultimately dominate the durability-favouring region of the front.

4) *Scope and limitations of the conclusions:* The scope of the conclusions should be kept explicit. First, all results are simulation based and the degradation objective is a trajectory-averaged control-oriented proxy rather than a measured capacity-fade endpoint. Second, the five-seed main comparison establishes performance relative to the selected comparator set; matched qEHVI/qNEHVI experiments are not included and the results should not be interpreted as an exhaustive state-of-the-art MOBO ranking. Third, the regional-guidance ablation contains only three matched seeds, so its independent contribution is less firmly established than that of LLM-informed initialization and is positioned as a secondary early-search refinement rather than as a peer contribution. Fourth, a controlled comparison across LLM backends and prompt realizations is outside the present study. Two supporting numerical diagnostics—Pareto-archive growth and objective-preprocessing sensitivity—are therefore retained in Appendix B, while broader MOBO baselines, cross-LLM studies, and physical-cell validation are left for future work.

VI. CONCLUSION

This paper presented LLMBO-MO, an LLM-guided extension of a ParEGO-style decomposition-based multiobjective Bayesian optimization framework for battery fast-charging protocol design under a limited simulation budget. The numerical MOBO backbone performs trade-off decomposition, GP surrogate modeling, expected-improvement optimization, simulator evaluation, and Pareto updating. LLM knowledge is introduced only through two search mechanisms: a protocol-domain-screened and diversity-aware initialization, and a weight-conditioned regional mechanism that supplies additional acquisition restarts during an early optimization window. All selected protocols are evaluated by the battery simulator before they are used for surrogate fitting or Pareto updating.

Under 56 simulator evaluations and five seeds, LLMBO-MO attains a mean final HV of 0.3848 ± 0.0073 on Chen2020, compared with 0.3520 ± 0.0169 for ParEGO, corresponding to a 9.3% improvement. A similar final-HV advantage is observed under the Ecker2015 parameterization. The matched initialization study provides the clearest mechanism-level evidence: battery-specific LLM information improves early search when simulator observations are scarce. The three-seed component ablation does not establish an independent systematic benefit from regional guidance, although the combined LLMBO-MO variant achieves the highest observed mean HV among the four short-budget variants. These results indicate that LLM knowledge is most reliably useful as early-stage search guidance within a numerical MOBO process.

Despite these results, several limitations should be kept in mind. First, the evaluation is simulation based: the SPM electrochemical and lumped thermal model in Section III captures the dominant dynamics of lithium-ion fast charging but does not resolve localized degradation, lithium plating, or thermal gradients, and the conclusions therefore do not transfer directly to physical cells. Second, the third objective

is a control-oriented trajectory-averaged degradation proxy rather than a measured capacity-fade endpoint, so the Pareto protocols should be interpreted in proxy units rather than as quantitative cycle-life predictions. Third, the five-seed main comparison is descriptive: the reported means and standard deviations characterize the observed seed distribution, and the relative improvement over ParEGO is small enough that an additional inferential test is needed before the difference should be called statistically reliable (paired test statistics are reported in Section V). Fourth, the three-seed regional-guidance ablation is statistically inconclusive, and the regional mechanism should therefore be read as a secondary early-search refinement rather than as a peer contribution to the LLM-informed initialization. Fifth, the present results are obtained with a single LLM backend (V4-Flash) on a single battery chemistry family; a controlled cross-LLM robustness study and additional parameterizations are required before the method can be claimed to be LLM-backend-agnostic or chemistry-agnostic. Finally, the Ecker2015 experiment is interpreted as a second-battery parameterization check rather than as a formal cross-battery transfer study, because the two parameterizations share the same simulator family and the same five seeds.

Future work will extend the evaluation to stronger matched MOBO baselines (notably qEHVI and qNEHVI), controlled comparisons across LLM backends and prompt realizations, broader battery operating conditions, and physical-cell experiments with experimentally grounded degradation endpoints.

APPENDIX A

OPTIONAL BOUNDED PREFERENCE COUPLING

Accepted regional anchors can additionally be translated into a bounded adjustment of the acquisition mean. This posterior-mean coupling is a diagnostic variant that is separate from the acquisition-restart regional guidance used by the principal LLMBO-MO configuration; it is therefore analyzed independently here.

Let $\mathcal{G}_t = \{\mathbf{g}_{j,t}\}_{j=1}^M$ denote accepted anchors. Each anchor is scored using its EI value, posterior uncertainty, and novelty:

$$\begin{aligned} n_{j,t} &= \min_{\mathbf{x} \in \mathcal{D}_t} \|\widehat{\mathbf{g}}_{j,t} - \widehat{\mathbf{x}}\|_2, \\ s_{j,t} &= \eta_{\text{EI}z}(\log \max\{\text{EI}_{j,t}, \epsilon_{\text{EI}}\}) + \eta_{\sigma z}(\sigma_{j,t}) + \eta_n z(n_{j,t}), \\ v_{j,t} &= \frac{\exp(s_{j,t}/T_v)}{\sum_{\ell=1}^M \exp(s_{\ell,t}/T_v)}. \end{aligned} \quad (14)$$

If anchor scoring is unavailable numerically, uniform weights $v_{j,t} = 1/M$ are used.

Let $\Sigma_{GG,t}$ be the posterior covariance among anchors and $\Sigma_{\theta G,t}$ the candidate-to-anchor covariance. The posterior correlation between a candidate and the weighted anchor portfolio is

$$\begin{aligned} V_{G,t} &= \mathbf{v}_t^\top (\Sigma_{GG,t} + \epsilon_K \mathbf{I}) \mathbf{v}_t, \\ \rho_t(\theta) &= \frac{\Sigma_{\theta G,t} \mathbf{v}_t}{\sigma_t(\theta) \sqrt{\max\{V_{G,t}, \epsilon_v\}}}. \end{aligned} \quad (15)$$

Only positive correlations are used to attract EI toward preferred regions.

The region-quality gate combines the concentration of anchor weights and the dispersion of anchor scores:

$$H_t = -\sum_{j=1}^M v_{j,t} \log(v_{j,t} + \varepsilon_H),$$

$$\bar{H}_t = \frac{H_t}{\max\{\log M, \varepsilon_H\}},$$

$$q_t = \text{clip}[\eta_H(1 - \bar{H}_t) + (1 - \eta_H) \tanh(\text{std}(s_t)), 0, 1].$$
(16)

The LLM-reported confidence $c_t = \text{clip}(h_t^{\text{LLM}}, 0, 1)$ and a bounded trust factor $\tau_t \in [0, 1]$ define $r_t = c_t \tau_t q_t$. The attenuation $a_t = \text{clip}(1 - (t-1)/T_L, 0, 1)$ reduces the possible influence to zero by the end of the guidance window. The bounded adjustment is

$$\delta_t(\theta) = \text{clip}(a_t \lambda_{\max} r_t [\rho_t(\theta)]_+, 0, \delta_{\max}),$$

$$\mu_t^L(\theta) = \mu_t(\theta) - \delta_t(\theta),$$

$$(\sigma_t^L)^2(\theta) = \sigma_t^2(\theta).$$
(17)

Because (9) is minimized, decreasing the posterior mean near positively correlated anchors increases their attractiveness to EI. Clipping limits the adjustment in standardized scalar-target units, preserves the posterior variance, and creates no pseudo-observation. The principal results use regional information only as acquisition-search restarts and therefore do not rely on (17).

APPENDIX B SUPPORTING NUMERICAL DIAGNOSTICS

A. Pareto-Archive Growth

The archive-growth diagnostic tracks the number of feasible Pareto-nondominated protocols retained during optimization. It describes archive richness and is not used as a substitute for HV or as a primary sample-efficiency metric. Here, an “optimal charging protocol” means a feasible Pareto-nondominated entry in the current archive.

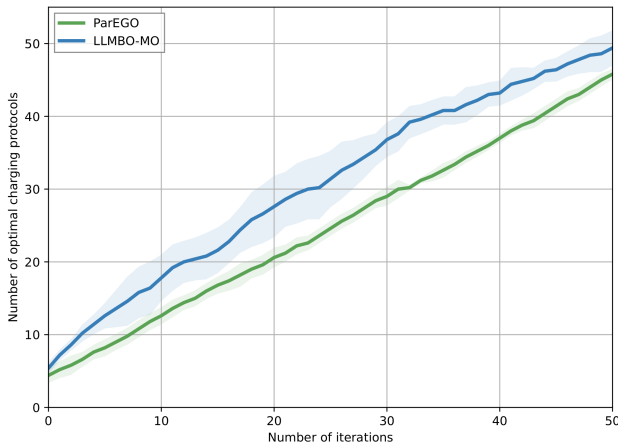


Fig. 7. Chen2020 Pareto-archive growth over 50 algorithmic iterations. Lines are five-seed LLMBO-MO and bands show across-seed dispersion.

At iteration 50, LLMBO-MO maintains an average of approximately 49.4 nondominated protocols, compared with 45.8 for ParEGO. Its curve remains higher through most of

TABLE VIII
OBJECTIVE-PREPROCESSING SENSITIVITY ON CHEN2020

Preprocessing	Evaluation 30	Evaluation 56
Min-max	0.3355	0.4146 $\pm$ 0.0067
Z-score	0.3920	0.4122 $\pm$ 0.0078
None	0.3255	0.3826 $\pm$ 0.0358

the search, indicating a richer retained set of trade-offs across these seeds. Because the horizontal axis counts algorithmic iterations, Fig. 7 is not an equal-simulator-call sample-efficiency comparison and is independent of the fixed-budget HV result.

B. Objective-Scaling Sensitivity

The scaling diagnostic compares min-max scaling, Z-score standardization, and no normalization on Chen2020 over five seeds and the same 56-evaluation budget. It tests numerical robustness rather than the LLM mechanism.

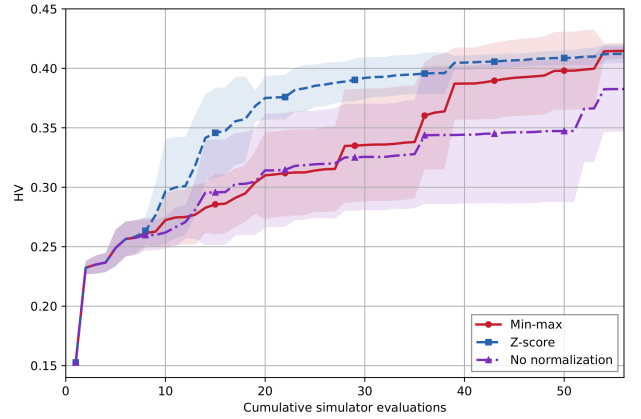


Fig. 8. Chen2020 objective-preprocessing sensitivity over five seeds. Lines show mean HV and bands show one sample standard deviation.

Z-score preprocessing progresses faster in the middle of the search, reaching an HV of 0.3920 at evaluation 30 compared with 0.3355 for min-max scaling. At evaluation 56, min-max scaling has the slightly higher mean (0.4146 versus 0.4122), but the difference is small and is not interpreted as a significant ordering. Both scale-balancing choices have substantially lower final dispersion than the unnormalized condition, whose final result is 0.3826 $\pm$ 0.0358. Thus, heterogeneous objective magnitudes should be balanced, while either conventional scaling choice is defensible in the present setting.

APPENDIX C SEED CONFIGURATION AND REPRODUCIBILITY

The main Chen2020 and Ecker2015 comparisons use the same five seeds (8409–8413) for each method. The reported studies share a common configuration:

- Chen2020 main comparison: five seeds (8409–8413); 56 simulator calls per seed.
- Ecker2015 second-parameterization case: five seeds (8409–8413); 56 simulator calls per seed.

- Initialization study: ten matched seeds; 26-call budget.
- Component ablation: three matched seeds (8409–8411); 16-call budget.
- Main regional guidance: $T_L = 16$, 64 anchors, EI-search restarts.
- LLM records: prompt configuration and responses logged per seed.

Seed-level identifiers, LLM call records, and parsed responses are retained in the experiment logs rather than in the main parameter table so that the reported statistics can be traced to the corresponding seed set without interrupting the main experimental narrative.

For the main algorithm, regional information affects only acquisition-search restarts. The posterior-mean coupling in Appendix A is a separate diagnostic realization and is not used to define the principal results.

REFERENCES

- [1] N. Nitta *et al.*, “Li-ion battery materials: Present and future,” *Materials Today*, vol. 18, no. 5, pp. 252–264, 2015.
- [2] A. Tomaszewska *et al.*, “Lithium-ion battery fast charging: A review,” *eTransportation*, vol. 1, Art. no. 100011, 2019.
- [3] X.-G. Yang and C.-Y. Wang, “Understanding the trilemma of fast charging, energy density and cycle life of lithium-ion batteries,” *J. Power Sources*, vol. 402, pp. 489–498, 2018.
- [4] B. Shahriari *et al.*, “Taking the human out of the loop: A review of Bayesian optimization,” *Proc. IEEE*, vol. 104, no. 1, pp. 148–175, 2016.
- [5] P. M. Attia *et al.*, “Closed-loop optimization of fast-charging protocols for batteries with machine learning,” *Nature*, vol. 578, pp. 397–402, 2020.
- [6] B. Jiang and X. Wang, “Constrained Bayesian optimization for minimum-time charging of lithium-ion batteries,” *IEEE Control Syst. Lett.*, vol. 6, pp. 1682–1687, 2022.
- [7] B. Jiang *et al.*, “Fast charging design for lithium-ion batteries via Bayesian optimization,” *Appl. Energy*, vol. 307, Art. no. 118244, 2022.
- [8] X. Wang and B. Jiang, “Multi-objective optimization for fast charging design of lithium-ion batteries using constrained Bayesian optimization,” *J. Power Sources*, vol. 584, Art. no. 233602, 2023, doi:10.1016/j.jpowsour.2023.233602.
- [9] B.-C. Wang, Y.-Y. Mao, Y. Wang, and H.-X. Li, “Gaussian process-accelerated multiobjective evolutionary design of charging process considering multiple user preferences,” *IEEE Trans. Ind. Informat.*, vol. 20, no. 8, pp. 10123–10133, 2024, doi:10.1109/TII.2024.3388602.
- [10] B.-C. Wang, R. Wang, Y.-Y. Mao, and H.-X. Li, “Experience-informed multiobjective optimization for fast charging design of lithium-ion batteries,” *IEEE Trans. Transp. Electrification*, 2026, doi:10.1109/TTE.2026.3610481.
- [11] M. F. Tahir, H. A. Khan, R. Asghar, and M. A. Zamora, “Multistage constant-current charging strategies for lithium-ion batteries: A review and future trends,” *J. Energy Chem.*, vol. 82, pp. 228–241, 2023, doi:10.1016/j.jechem.2023.03.022.
- [12] W. Jeong, S. Heo, and J. H. Lee, “Health-conscious charging of lithium-ion batteries using Bayesian optimization guided by a semi-empirical aging model,” *J. Energy Storage*, vol. 141, Art. no. 119348, 2026.
- [13] G. Dong, Y. Feng, Y. Lou, M. Zhang, and J. Wei, “Data-driven fast charging optimization for lithium-ion battery using Bayesian optimization with fast convergence,” *IEEE Trans. Transp. Electrification*, vol. 10, no. 2, pp. 4173–4183, 2024.
- [14] G. Luo, D. Han, Y. Zhang, and H. Ruan, “A digital twin for advancing battery fast charging based on a Bayesian optimization-based method,” *J. Energy Storage*, vol. 93, Art. no. 112365, 2024.
- [15] X. Song and B. Jiang, “Parallel Bayesian optimization using satisficing Thompson sampling for fast charging design of lithium-ion batteries,” *Eng. Appl. Artif. Intell.*, vol. 154, Art. no. 110868, 2025.
- [16] R. Zhu, W. Peng, F. Yang, and M. Xie, “Fast-charging protocols design of lithium-ion battery: A multiple-objective Bayesian optimization perspective,” *IEEE Trans. Transp. Electrification*, vol. 11, no. 3, pp. 8327–8338, 2025.
- [17] S. Yoon, Y. Lee, and H.-K. Kim, “Enhancing fast-charging protocols with section-based Bayesian optimization for lithium-ion batteries to prevent Li-plating,” *eTransportation*, vol. 26, Art. no. 100460, 2025.
- [18] C. Peng, S. Yan, C. Zhong, Q. Huang, C. Wu, and H. Huang, “Learning-based temporal sequence of constrained handling selection for constrained multi-objective evolutionary optimization,” *IEEE Trans. Evol. Comput.*, vol. 30, no. 3, pp. 1123–1136, 2026, doi:10.1109/TEVC.2025.3584207.
- [19] C. Peng, J. Zhang, Z. Cai, Y. Xu, Q. Huang, and H. Huang, “Strategy selection in dynamic constrained multi-objective optimization via state-augmented deep reinforcement learning,” *IEEE Trans. Evol. Comput.*, early access, 2026, doi:10.1109/TEVC.2026.3678702.
- [20] J. Knowles, “ParEGO: A hybrid algorithm with on-line landscape approximation for expensive multiobjective optimization problems,” *IEEE Trans. Evol. Comput.*, vol. 10, no. 1, pp. 50–66, 2006.
- [21] Z. Zhang, Y. Wang, G. Sun, and T. Pang, “A distribution information-based Kriging-assisted evolutionary algorithm for expensive many-objective optimization problems,” *IEEE Trans. Evol. Comput.*, vol. 29, no. 6, pp. 2656–2670, 2025.
- [22] Y. Li, W. Li, S. Li, and Y. Zhao, “An infill sampling criterion based on improvement of probability and mapping crowding distance for expensive multi/many-objective optimization,” *Eng. Appl. Artif. Intell.*, vol. 133, Art. no. 108616, 2024.
- [23] T. Liu, N. Astorga, N. Seedat, and M. van der Schaar, “Large language models to enhance Bayesian optimization,” in *Proc. Int. Conf. Learn. Represent.*, 2024, [Online].
- [24] X. Yuan, Z. Chen, J. Zhang, H. Xiong, N. Ye, Y. Li, and Q. Gu, “Unleashing LLMs in Bayesian optimization: Preference-guided framework for scientific discovery,” *arXiv preprint arXiv:2605.17976*, 2026, arXiv:2605.17976.
- [25] Z. Chen *et al.*, “LABO: LLM-accelerated Bayesian optimization through broad exploration and selective experimentation,” *arXiv preprint arXiv:2605.22054*, 2026, arXiv:2605.22054.
- [26] X. Kuai, J.-H. Ren, B.-C. Wang, and Y. Feng, “Large language model-enhanced Bayesian optimization for parameter identification of lithium-ion batteries,” *J. Energy Storage*, vol. 135, Art. no. 118198, 2025, doi:10.1016/j.est.2025.118198.
- [27] A. Kristiadi, F. Strieth-Kalthoff, M. Skreta, P. Poupart, A. Aspuru-Guzik, and G. Pleiss, “A sober look at LLMs for material discovery: Are they actually good for Bayesian optimization over molecules?” in *Proc. 41st Int. Conf. Mach. Learn.*, PMLR, vol. 235, pp. 25603–25622, 2024, [Online].
- [28] G. Suri and S. Onori, “A control-oriented cycle-life model for hybrid electric vehicle lithium-ion batteries,” *Energy*, vol. 96, pp. 644–653, 2016.
- [29] S. G. Marquis, V. Sulzer, R. Timms, C. P. Please, and S. J. Chapman, “An asymptotic derivation of a single particle model with electrolyte,” *J. Electrochem. Soc.*, vol. 166, no. 15, pp. A3693–A3706, 2019.
- [30] V. Sulzer *et al.*, “Python Battery Mathematical Modelling (PyBaMM),” *J. Open Res. Softw.*, vol. 9, no. 1, Art. no. 14, 2021.
- [31] C.-H. Chen, F. Brosa Planella, K. O’Regan, D. Gastol, W. D. Widanage, and E. Kendrick, “Development of experimental techniques for parameterization of multi-scale lithium-ion battery models,” *J. Electrochem. Soc.*, vol. 167, no. 8, Art. no. 080534, 2020.
- [32] M. Ecker, T. K. D. Tran, P. Dechent, S. Käbitz, A. Warnecke, and D. U. Sauer, “Parameterization of a physico-chemical model of a lithium-ion battery: I. Determination of parameters,” *J. Electrochem. Soc.*, vol. 162, no. 9, pp. A1836–A1848, 2015.
- [33] J. G. Falcón-Cardona, H. Ishibuchi, and C. A. Coello Coello, “Riesz s -energy-based reference sets for multi-objective optimization,” in *Proc. IEEE Congr. Evol. Comput.*, pp. 1–8, 2020.
- [34] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, “A fast and elitist multiobjective genetic algorithm: NSGA-II,” *IEEE Trans. Evol. Comput.*, vol. 6, no. 2, pp. 182–197, 2002.